File Handling in Python

File handling allows a program to read data from files and write data to files.
Python provides built-in functions and methods to work with files easily.

Opening a File

Files are opened using the open() function.

file = open("data.txt", "r")

The first argument is the file name. The second argument is the file mode.

File Modes

Commonly used file modes:

Mode	**Description**
r	Read mode
w	Write mode
a	Append mode
x	Create file
rb	Read binary
wb	Write binary

Reading a File

read()

Reads the entire file content as a string.

file = open("data.txt", "r")

content = file.read()

print(content)

file.close()

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readline()

Reads one line at a time.

```
file = open("data.txt", "r")
line = file.readline()
print(line)
file.close()
```

---

readlines()

Reads all lines and returns them as a list.

```
file = open("data.txt", "r")
lines = file.readlines()
file.close()
```

Writing to a File

Write Mode (w**)**

Creates a new file or overwrites an existing file.

```
file = open("data.txt", "w")
file.write("Hello Python")
file.close()
```

Append Mode (a**)**

Adds content to the end of a file.

```
file = open("data.txt", "a")
file.write("\nNew line added")
file.close()
```

Using the `with` Statement

The with statement automatically closes the file.

```
with open("data.txt", "r") as file:
   content = file.read()
   print(content)
```

This is the recommended way to work with files.

Writing Multiple Lines

```
lines = ["Line 1\n", "Line 2\n", "Line 3\n"]


with open("data.txt", "w") as file:
   file.writelines(lines)
```

Checking File Existence

```
import os


if os.path.exists("data.txt"):
   print("File exists")
```

Reading and Writing Binary Files

```
with open("image.png", "rb") as file:
   data = file.read()
with open("copy.png", "wb") as file:
   file.write(data)
```